

NIVELL 1: Creació BBDD

Creo la BBDD anomenada SPRINT5 i dins d'elles creo 5 col·leccions, una per cada fitxer .json que importo.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'IT-ACADEMY' database with collections: comments, movies, sessions, theaters, and users. The 'users' collection is selected. The main panel shows the 'Documents' tab with 185 documents. A query bar at the top contains '{ field: 'value' }' and a 'Generate query' button. Below the query bar, there are buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', and 'DELETE'. A dropdown menu is open, showing options to 'Import JSON or CSV file' and 'Insert document'. A notification at the bottom left states 'Import completed. 185 documents imported.'

comments

Storage size: 6.93 MB
Documents: 50 K
Avg. document size: 284.00 B
Indexes: 1
Total index size: 524.29 kB

movies

Storage size: 21.61 MB
Documents: 24 K
Avg. document size: 1.60 kB
Indexes: 1
Total index size: 585.73 kB

sessions

Storage size: 20.48 kB
Documents: 1
Avg. document size: 540.00 B
Indexes: 1
Total index size: 20.48 kB

theaters

Storage size: 147.46 kB
Documents: 1.6 K
Avg. document size: 223.00 B
Indexes: 1
Total index size: 32.77 kB

users

Storage size: 36.86 kB
Documents: 185
Avg. document size: 159.00 B
Indexes: 1
Total index size: 20.48 kB

NIVELL 1: Exercici 1

Mostra els 2 primers comentaris que hi ha en la base de dades.

Opció1: des de la Interface del Compass. Ordeno per data asc i limito a 2 els registres.

IT-ACADEMY > SPRINT5 > comments [Open MongoDB shell](#)

Documents 50.3K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

Project { field: 0 }

Sort {date:1}

Collation { locale: 'simple' }

Index Hint { field: -1 }

Max Time MS 60000

Skip 0 Limit 2

[ADD DATA](#) [EXPORT DATA](#) [UPDATE](#) [DELETE](#) 25 1 - 2 of 2

```
{
  "_id": ObjectId("5a9427648b0beebe695e67e"),
  "name": "Mercedes Tyler",
  "email": "mercedes_tyler@fakegmail.com",
  "movie_id": ObjectId("573a1399f29313caabcec3ce"),
  "text": "Optio totam dolores magni. Enim ratione fuga tempora voluptatum est cu...",
  "date": 1970-01-01T01:07:09.000+00:00
}
```

```
{
  "_id": ObjectId("5a9427658b0beebe6969efe"),
  "name": "Jon Snow",
  "email": "kit_harrington@gameofthron.es",
  "movie_id": ObjectId("573a13a4f29313caabd117df"),
  "text": "Dolorem animi tempora ullam quas. Iusto nobis reprehenderit aspernatur...",
  "date": 1970-01-01T09:37:49.000+00:00
}
```

Opció 2: des del Shell

```
> db.comments.find().sort({date:1}).limit(2)
< {
  "_id": ObjectId("5a9427648b0beebe695e67e"),
  "name": "Mercedes Tyler",
  "email": "mercedes_tyler@fakegmail.com",
  "movie_id": ObjectId("573a1399f29313caabcec3ce"),
  "text": "Optio totam dolores magni. Enim ratione fuga tempora voluptatum est cumque animi. Quia nam doloribus corrupti id voluptat",
  "date": 1970-01-01T01:07:09.000Z
}
{
  "_id": ObjectId("5a9427658b0beebe6969efe"),
  "name": "Jon Snow",
  "email": "kit_harrington@gameofthron.es",
  "movie_id": ObjectId("573a13a4f29313caabd117df"),
  "text": "Dolorem animi tempora ullam quas. Iusto nobis reprehenderit aspernatur cupiditate a tempore. Commodi alias expedita dolor",
  "date": 1970-01-01T09:37:49.000Z
}
SPRINT5>
```

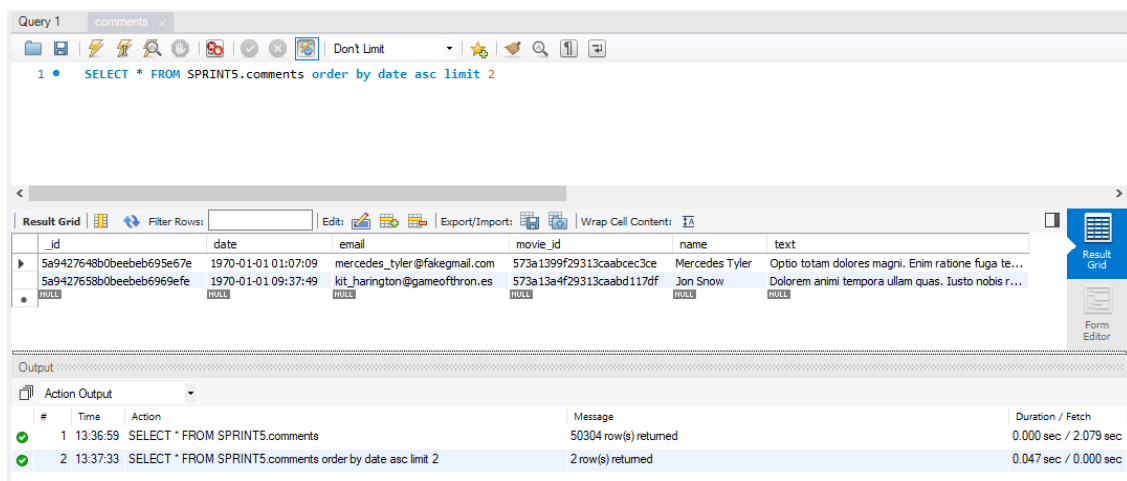
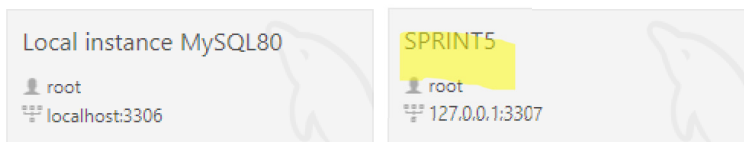
Opció 3: des del MySQL

Creo una connexió al *MySQL* executant el driver/connector *mogosqld* per apuntar a la base de dades de *mongoDB*.

```
C:\> Indicador d'ordres - mongosqld.exe --config C:\mongodb\mongosqld-config.yaml
```

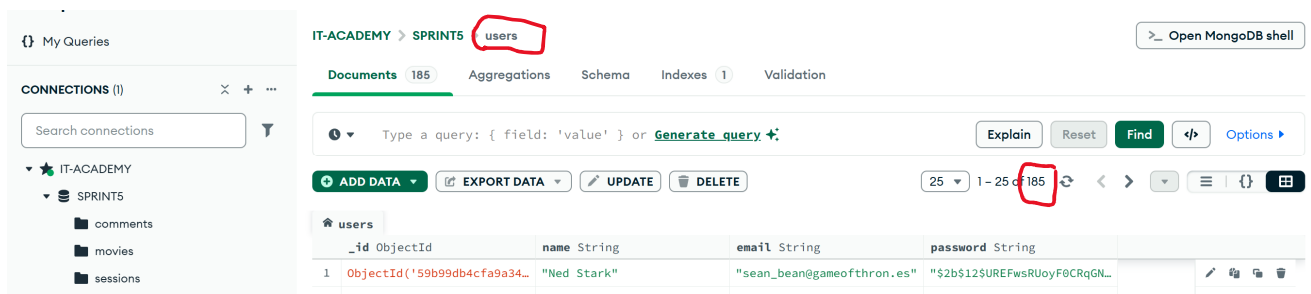
```
C:\Program Files\MongoDB\Connector for BI\2.14\bin>mongosqld.exe --config C:\mongodb\mongosqld-config.yaml
```

MySQL Connections +



Quants usuaris tenim registrats?

Opció1: des de la Interface del Compass mostra el total de registres.



Opció 2: des del Shell

```
>_MONGOSH

> use SPRINT5
< switched to db SPRINT5
> db.users.countDocuments()
< 185
SPRINT5>
```

Opció 3: des del MySQL

Creo una connexió al *MySQL* executant el driver/connector *mogosqld* per apuntar a la base de dades de *mongoDB*.

```
C:\Program Files\MongoDB\Connector for BI\2.14\bin>mongosqld.exe
2025-07-17T11:00:09.709+0200 I CONTROL [initandlisten] mongosqld starting: version=v2.14.24 pid=36012 host=ELITEX2-PC
2025-07-17T11:00:09.749+0200 I CONTROL [initandlisten] git version: 5fc912c605f3aadb8c75157d109b74de2cddd1f9
2025-07-17T11:00:09.749+0200 I CONTROL [initandlisten] OpenSSL version OpenSSL 3.5.0 8 Apr 2025 (built with OpenSSL 3.5.0 8 Apr 2025)
2025-07-17T11:00:09.749+0200 I CONTROL [initandlisten] options: {}
2025-07-17T11:00:09.749+0200 I CONTROL [initandlisten] ** WARNING: Access control is not enabled for mongosqld.
2025-07-17T11:00:09.749+0200 I CONTROL [initandlisten]
2025-07-17T11:00:09.752+0200 I NETWORK [initandlisten] waiting for connections at 127.0.0.1:3307
2025-07-17T11:00:09.767+0200 I SCHEMA [sampler] sampling MongoDB for schema...
2025-07-17T11:00:10.113+0200 I SCHEMA [sampler] mapped schema for 5 namespaces: "SPRINT5" (5): ["comments", "sessions", "theaters", "movies", "users"]
2025-07-17T11:00:43.281+0200 I NETWORK [conn1] connection accepted from 127.0.0.1:60061 #1 (1 connection now open)
2025-07-17T11:00:43.517+0200 I NETWORK [conn2] connection accepted from 127.0.0.1:60066 #2 (2 connections now open)
```

Query 1 | comments | users | Don't Limit

```
1 • SELECT count(*) FROM SPRINT5.users;
```

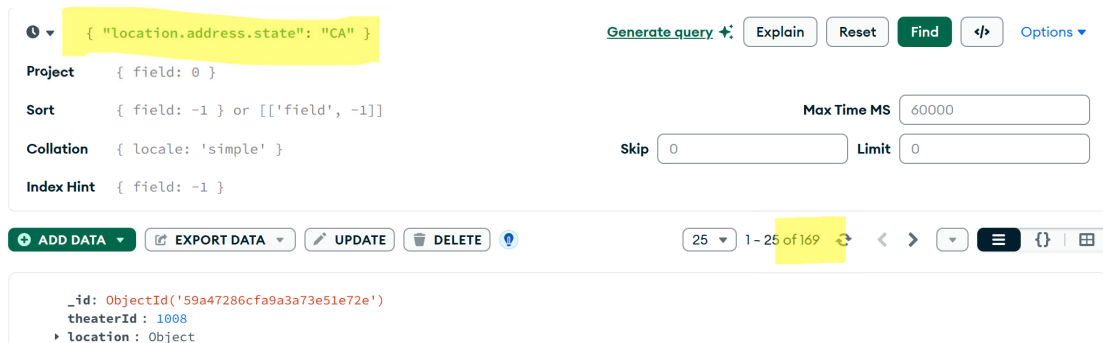
count(*)
185

Output

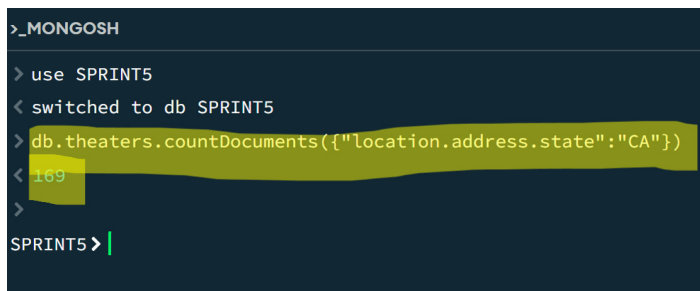
#	Time	Action	Message	Duration / Fetch
✓ 1	11:00:58	SELECT * FROM SPRINT5.users	185 row(s) returned	0.016 sec / 0.000 sec
✓ 2	11:01:09	SELECT count(*) FROM SPRINT5.users	1 row(s) returned	0.016 sec / 0.000 sec

Quants cinemes hi ha en l'estat de Califòrnia?

Opció1: des de la Interface del Compass. Ordeno per data asc i limito a 2 els registres.

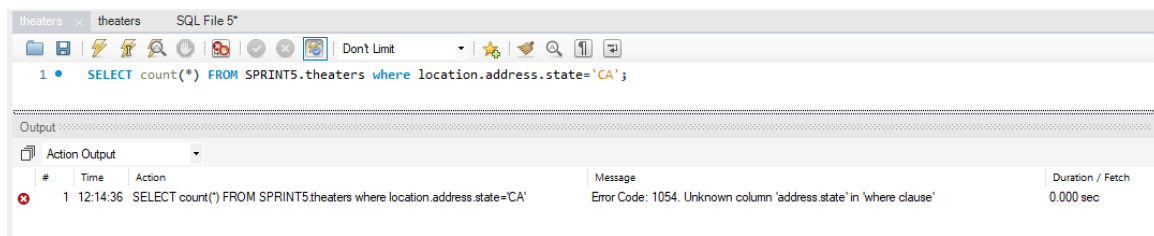


Opció 2: des del Shell



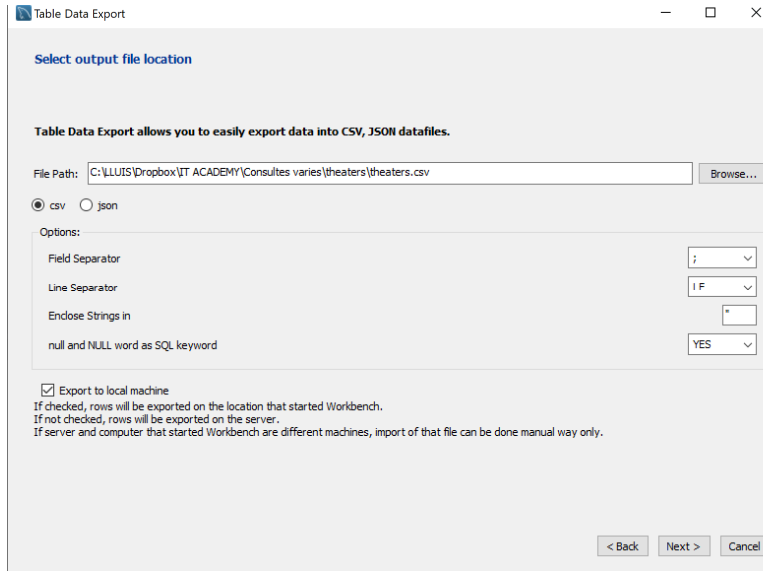
Opció 3: des del MySql

No es pot fer una consulta directe sobre el camp State amb driver/connectors *mongoose* perquè les dades que volem consultar provenen d'una **estructura JSON o anidada**, el que no és vàlid en columnes normals.

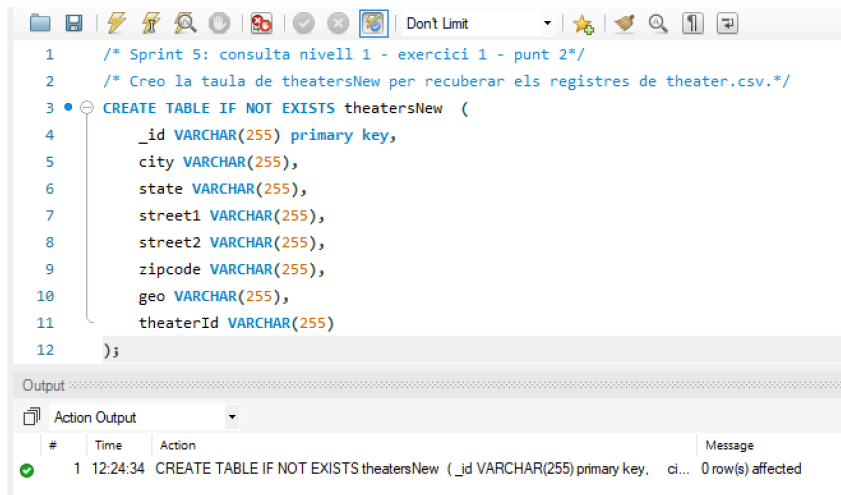


ALTERNATIVA: Exportat la taula a .csv i després importar-la en una nova taula.

1-Exporto la taula a un fitxer *theaters.csv* mitjançant el Wizard.



2- Obro una nova connexió al MySQL i creo una taula nova *theatersNew* dins de la BBDD SPRINT4 amb els camps del fitxer *theaters.csv*:



3- Importo els registres des de *theaters.csv*:

The screenshot shows the MongoDB Compass interface. The top pane contains the following SQL command:

```
1 • LOAD DATA INFILE 'C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\theaters.csv'
2 INTO TABLE theatersnew
3 FIELDS TERMINATED BY ','
4 ENCLOSED BY ''
5 IGNORE 1 LINES;
6
7 • select * from theatersnew;
```

The bottom pane displays the 'Result Grid' with a table of imported data:

_id	city	state	street1	street2	zipcode	geo	theaterId
59a47286cfa9a3a73e51e72c	Bloomington	MN	340 W Market		55425	Point	1000
59a47286cfa9a3a73e51e72d	California	MD	45235 Worth Ave.		20619	Point	1003
59a47286cfa9a3a73e51e72e	Vacaville	CA	1621 E Monte Vista Ave		95688	Point	1008
59a47286cfa9a3a73e51e72f	Birmingham	AL	5072 Pinnacle Sq		35235	Point	1004
59a47286cfa9a3a73e51e730	Tempe	AZ	1900 E Rio Salado Pkwy		85281	Point	1002
59a47286cfa9a3a73e51e731	Clarksville	IN	1025 Veterans Pkwy		47129	Point	1010

The 'Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	12:29:07	LOAD DATA INFILE C:\\ProgramData\\MySQL\\MySQL Server 8.0\\Uploads\\the...	1564 row(s) affected Records: 1564 Deleted: 0 Skipped: 0 Warnings: 0	0.125 sec
2	12:29:12	select * from theatersnew	1564 row(s) returned	0.000 sec / 0.000 sec

4- Faig la consulta “Quants cinemes hi ha en l'estat de Califòrnia?”

The screenshot shows the MongoDB Compass interface with the following SQL query:

```
1 • SELECT count(*) FROM sprint4.theatersnew where state = 'CA';
```

The 'Result Grid' shows the result of the query:

count(*)
169

The 'Output' pane shows the execution log:

#	Time	Action	Message
1	12:32:45	SELECT count(*) FROM sprint4.theatersnew where state = 'CA'	1 row(s) returned

Quin va ser el primer usuari/ària en registrar-se?

Opció1: des de la Interface del Compass. Ordono per data asc i limito a 1 els registres. Mostrant només el nom i la data.

The screenshot shows the MongoDB Compass interface with the following query:

```
{ field: 'value' } or Generate query
```

The query is executed, and the result is displayed in the 'Result Grid':

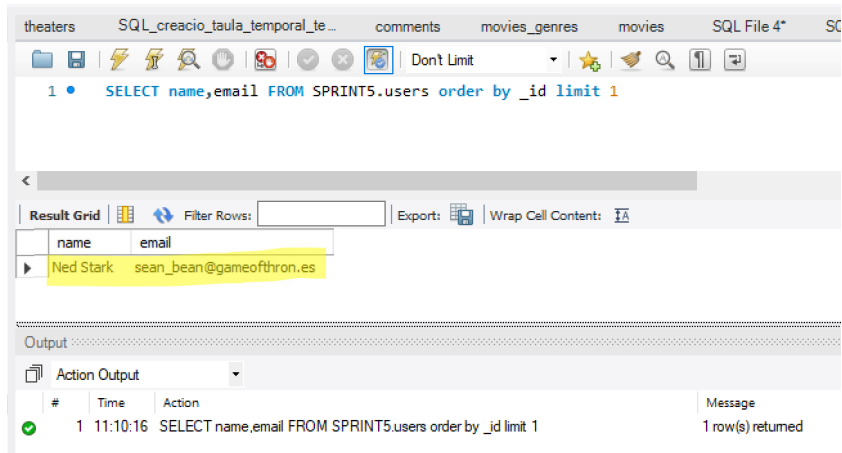
_id	name	email	password
ObjectId('59b99db4cfa9a34dcd7885b6')	Ned Stark	sean_bean@gameofthron.es	\$2b\$12\$UREFwsUoyF0CRqGNK0Lz06HM/jLhgUCNNI39R3AqMUQ74crl31Vu

Opció 2: des del Shell

```
>_MONGOSH

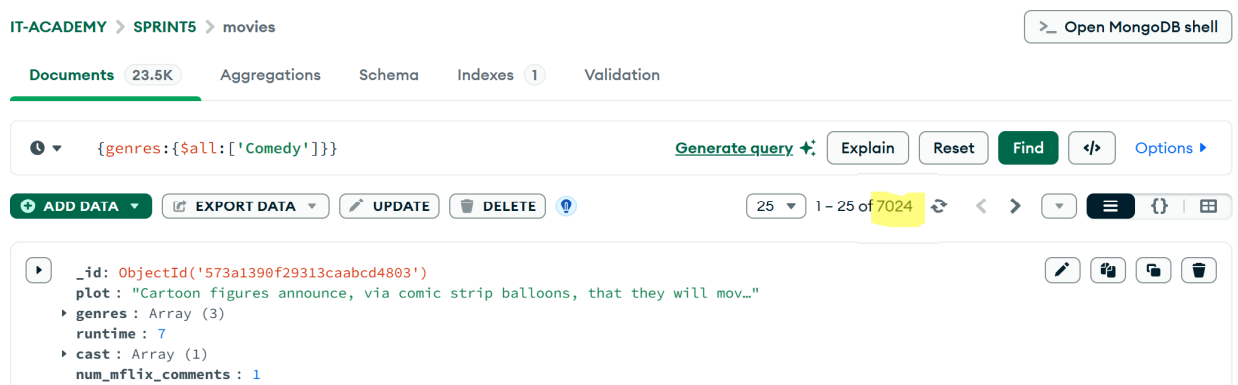
> use SPRINT5
< switched to db SPRINT5
> db.users.find().sort({_id:1}).limit(1)
< {
  _id: ObjectId('59b99db4cf9a34dcd7885b6'),
  name: 'Ned Stark',
  email: 'sean_bean@gameofthron.es',
  password: '$2b$12$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNIJ9RJAqMUQ74cr1J1Vu'
}
SPRINT5> |
```

Opció 3: des del MySql



Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

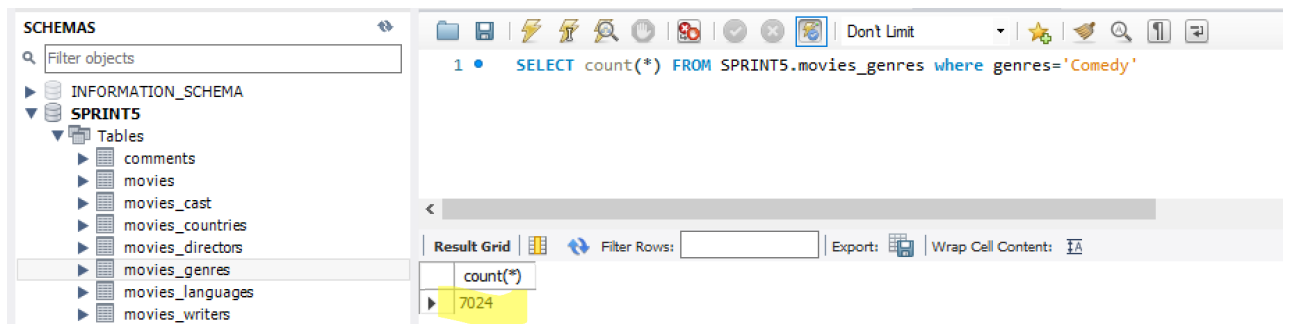
Opció1: des de la Interface del Compass. Faig el filtre { genres: ["Comedy"] }



Opció 2: des del Shell

```
>_MONGOSH
> db.movies.find({"genres":"Comedy"}).count()
< 7024
> db.movies.countDocuments({"genres":"Comedy"})
< 7024
> db.movies.countDocuments({genres:'Comedy'})
< 7024
> db.movies.countDocuments({genres:{$all:["Comedy"]}})
< 7024
SPRINT5> |
```

Opció 3: des del MySQL.



The screenshot shows a MySQL database interface. On the left, the 'SCHEMAS' panel is open, showing a tree view with 'INFORMATION_SCHEMA' and 'SPRINT5'. Under 'SPRINT5', there is a 'Tables' folder containing several tables: 'comments', 'movies', 'movies_cast', 'movies_countries', 'movies_directors', 'movies_genres', 'movies_languages', and 'movies_writers'. The 'movies_genres' table is selected. On the right, the SQL editor shows a query: `SELECT count(*) FROM SPRINT5.movies_genres where genres='Comedy'`. Below the query, the 'Result Grid' shows a single row with the value '7024' under the column 'count(*)'.

NIVELL 1: Exercici 2

Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès.

Opció 1: Des del Compass

T-ACADEMY > SPRINT5 > movies [Open MongoDB shell](#)

Documents 23.5K Aggregations Schema Indexes 1 Validation

```
{
  year: 1932,
  $or: [
    { genres: { $all: ["Drama"] } },
    { languages: { $all: ["French"] } }
  ]
}
```

Project {title:1,_id:0}

Sort {title:1}

Collation { locale: 'simple' }

Index Hint { field: -1 }

Max Time MS 60000

Skip 0 Limit 0

EXPORT DATA 25 1-18 of 18

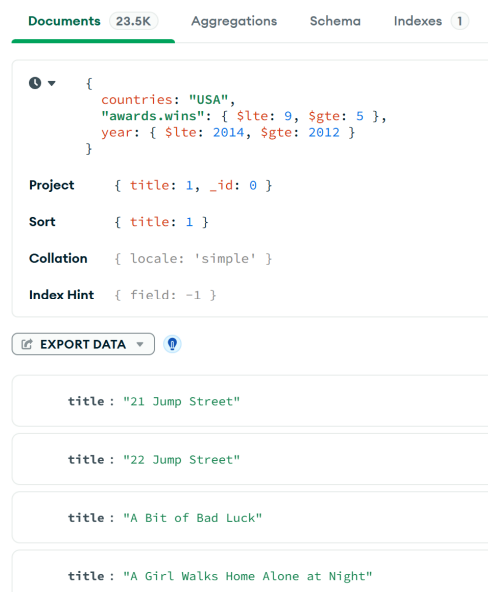
	title String
1	"A Farewell to Arms"
2	"Broken Lullaby"
3	"Forbidden"
4	"Freaks"
5	"Grand Hotel"

Opció 2: des del Shell

```
>_MONGOSH
>
> db.movies.find({year:1932,$or:[{genres:'Drama'},{languages:'French'}]},{title:1,_id:0}).sort({title:1})
< {
  title: 'A Farewell to Arms'
}
{
  title: 'Broken Lullaby'
}
{
  title: 'Forbidden'
}
{
  title: 'Freaks'
}
{
  title: 'Grand Hotel'
}
{
  title: 'I Am a Fugitive from a Chain Gang'
}
```

NIVELL 1: Exercici 3

Mostra'm tots els documents de pel·lícules estaunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

Opció 1: des del Compass**Opció 2: des del Shell**

NIVELL 2: Exercici 1

Compte quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

Opció 1: des del Compass

The screenshot shows the MongoDB Compass web interface. At the top, there are tabs for Documents (185), Aggregations, Schema, Indexes (1), and Validation. Below the tabs, a query is entered in the search bar: `{ email: { $regex: 'gameofthron\\.es$' } }`. To the right of the query bar are buttons for 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'. Below the query bar, there are buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', and 'DELETE'. A status bar indicates '25' and '1 - 25 of 83'. The results are displayed in a list format, showing two documents:

```
{
  "_id": ObjectId('59b99db4cfa9a34dcd7885b6'),
  "name": "Ned Stark",
  "email": "sean_bean@gameofthron.es",
  "password": "$2b$12$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNIJ9RJAqMUQ74crLJ1Vu"
}
```

```
{
  "_id": ObjectId('59b99db4cfa9a34dcd7885b7'),
  "name": "Robert Baratheon",
  "email": "mark_addy@gameofthron.es",
  "password": "$2b$12$yGqxLG9LZpXA2xVDhuPnS0Zd.VURVxz7wg0LY3pn00s7u2S1Z032y"
}
```

Opció 2: des del Shell

```
> _MONGOSH
> db.users.countDocuments({ email: { $regex: 'gameofthron\\.es$' } })
< 83
SPRINT5 >
```

NIVELL 2: Exercici 2

Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

Opció 1: des del Compass

The screenshot shows the MongoDB Compass interface. The top navigation bar includes 'Documents' (1.6K), 'Aggregations', 'Schema', 'Indexes' (1), and 'Validation'. The main query editor shows a filter: `{ "location.address.state": "DC" }`. The project is set to `{ "location.address.zipcode": 1, theaterId: 1, _id: 0 }`. The sort is empty. The collation is empty. The index hint is `{ field: -1 }`. The 'Skip' field is set to 0 and the 'Limit' field is empty. The 'Max Time MS' field is empty. The 'EXPORT DATA' button is visible. The results section shows three documents:

```
theaterId : 1092
location : Object
  address : Object
    zipcode : "20010"

theaterId : 801
location : Object
  address : Object
    zipcode : "20016"

theaterId : 8500
location : Object
  address : Object
    zipcode : "20002"
```

Opció 2: des del Shell

```
> _MONGOSH
> db.theaters.aggregate([
  {
    $match: { "location.address.state": "DC" }
  },
  {
    $group: {
      _id: "$location.address.zipcode",
      totalTeatros: { $sum: 1 }
    }
  }
])
< {
  _id: '20002',
  totalTeatros: 1
}
{
  _id: '20010',
  totalTeatros: 1
}
{
  _id: '20016',
  totalTeatros: 1
}
SPRINT5>
```

NIVELL 3: Exercici 1

Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

Opció 1:

The screenshot shows the MongoDB Compass web interface. At the top, there's a breadcrumb trail: IT-ACADEMY > SPRINTS > movies. On the right, there's a button 'Open MongoDB shell'. Below the breadcrumb, there are tabs: Documents (23.5K), Aggregations, Schema, Indexes (1), and Validation. The 'Documents' tab is active. In the center, there's a query editor with the following JSON query:

```
{  "directors": { "$all": ["John Landis"] },  "imdb.rating": { "$lte": 8, "$gte": 7.5 } }
```

 To the right of the query editor are buttons: 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'. Below the query editor, there's an 'EXPORT DATA' button. To the right of the export button, there's a dropdown menu set to '25' and a pagination indicator '1 - 4 of 4'. Below these, there's a list of four movie documents. Each document is shown in a light blue box with expandable sections for 'imdb', 'title', and 'directors'. The movies listed are: 'Animal House' (rating 7.6), 'The Blues Brothers' (rating 7.9), 'An American Werewolf in London' (rating 7.6), and 'Trading Places' (rating 7.5).

He exportat el resultat de la consulta a un fitxer .csv i aquest és el resultat:

```
1  imdb.rating,title,directors[0]
2  7.6,Animal House,John Landis
3  7.9,The Blues Brothers,John Landis
4  7.6,An American Werewolf in London,John Landis
5  7.5,Trading Places,John Landis
```

Opció 2: des del Shell

```
> db.movies.find(
  {
    directors: { $all: ["John Landis"] },
    "imdb.rating": { $gte: 7.5, $lte: 8 }
  },
  {
    title: 1,
    directors: 1,
    "imdb.rating": 1,
    "_id": 0
  }
)
< {
  imdb: {
    rating: 7.6
  },
  title: 'Animal House',
  directors: [
    'John Landis'
  ]
}
{
  title: 'The Blues Brothers',
  directors: [
    'John Landis'
  ],
  imdb: {
    rating: 7.9
  }
}
```

NIVELL 3: Exercici 2

Mostra en un mapa la ubicació de tots els teatres de la base de dades.

